Concepts of Biostatistics and Data Science

Participant workbook for the two-day crash course with practical analysis in R

Neudata Consulting Ltd · Version 1.0 · 26-10-2026

Name: _____ Date: _____

How to use this workbook

This workbook accompanies the two-day crash course in the concepts of biostatistics and data science, and follows its practical sessions. Each exercise names the R script it belongs to. The scripts contain worked examples with comments; the exercises ask you to adapt that code to a new question and to interpret the result in words. Write your answers in the boxes.

- Core exercises (numbered 1.1, 2.1 ...) form the guided pathway. Everyone should aim to complete them.
- Extensions (E1.1, E2.1 ...) are optional, for participants who finish early or have more experience.
- Always write the interpretation, not only the number. A good answer gives the estimate, its 95% CI, and what it means.
- The reference section at the end contains the cheat sheet, decision diagrams and method-selection guide.

> The Healthy Hearts District Study is simulated. Its results are for teaching only and must not be quoted as findings about real people.

Getting started in RStudio

1. Open `Concepts-of-Biostatistics-and-Data-Science.Rproj` (double-click). This sets the working folder.
2. Open a script from the Files pane (bottom right), folder `R`.
3. Put the cursor on a line and press Ctrl+Enter (Cmd+Enter on a Mac) to run it. Results appear in the Console (bottom left), plots in the Plots pane.
4. Run lines in order from the top. If you get an error, read the message, check spelling and brackets, and check that earlier lines were run.
5. Type your own code at the bottom of each script, under "Your turn". Save with Ctrl+S.

Symbol	Meaning	Example
<-	Store a result under a name	`fit <- lm(sbp_0 ~ age, data = hearts)`
~	"is modelled by" / "by group"	`t.test(sbp_0 ~ sex, data = hearts)`
|>	Pass the result on to the next step	`hearts |> count(district)`
$	One column of a data frame	`hearts$age`
na.rm = TRUE	Ignore missing values	`mean(hearts$crp, na.rm = TRUE)`
?name	Open the help page	`?t.test`

The data

480 adults aged 30 to 74 from four districts were measured at baseline and randomly allocated to standard advice or a lifestyle programme. Blood pressure, satisfaction, adverse events and clinic visits were recorded over 12 months. Some participants left early.

Variable	Description	Type	Units or coding	Missing
id	Participant identifier	Identifier	HH001 to HH480	0
district	District of residence	Nominal (4 categories)	North; South; East; West	0
sex	Sex	Binary	Female; Male	0
age	Age at baseline	Continuous	Years (30 to 74)	0
education	Highest education level	Ordinal (3 categories)	Primary or less; Secondary; Tertiary	6
smoker	Current smoker at baseline	Binary	No; Yes	0
activity_min	Moderate physical activity	Continuous (right-skewed)	Minutes per week	7
diet_score	Healthy eating score	Discrete score (treated as continuous)	0 (least healthy) to 10 (most healthy)	0
bmi	Body mass index at baseline	Continuous	kg/m2	10
sbp_0	Systolic blood pressure at baseline	Continuous	mmHg	0
high_bp_0	High blood pressure at baseline	Binary	Yes if sbp_0 >= 140 mmHg; otherwise No	0
crp	C-reactive protein at baseline	Continuous (right-skewed)	mg/L	14
programme	Randomised group	Binary	Standard advice; Lifestyle programme	0
sbp_12	Systolic blood pressure at 12 months	Continuous	mmHg	41
high_bp_12	High blood pressure at 12 months	Binary	Yes if sbp_12 >= 140 mmHg; otherwise No	41
satisfaction	Satisfaction with care at 12 months	Ordinal (5 levels)	1 = very dissatisfied to 5 = very satisfied	41
adverse_event	Minor injury reported during follow-up	Binary	No; Yes	0
clinic_visits	Unplanned clinic visits during follow-up	Count	Number of visits	0
followup_months	Months of follow-up	Continuous (exposure time)	Months (2 to 12)	0

Day 1 practical: understanding data and making comparisons

Part A. Import and inspect (script 01_import_and_inspect.R, 25 minutes)

Exercise 1.1. Import the raw data

Run the first sections of the script to import `data/hearts_raw.csv`.

a) How many participants (rows) and variables (columns) are there?

Your answer

b) What does one row represent? Which two variables are measurements of the same thing on the same person at different times?

Your answer

Exercise 1.2. Find and fix missing values

Use `summary(raw)` and `colSums(is.na(raw))`.

a) Which variables have missing values, and how many?

Your answer

b) What problem do you see in `activity_min`? How does the script fix it? How many missing values does `activity_min` have after the fix?

Your answer

c) 41 people have no 12-month blood pressure. Why might this matter for the results?

Your answer

Exercise 1.3. Variable types

Classify each variable. Use: continuous, count, binary, nominal, ordinal.

Variable	Type	Variable	Type
district		satisfaction	
age		clinic_visits	
education		crp	
smoker		high_bp_0	
sbp_0		followup_months	

Part B. Describe and plot (script 02_describe_and_plot.R, 30 minutes)

Exercise 1.4. Counts and proportions

Use `count()` and `mutate()` as in the script.

```
count(hearts, smoker) |> mutate(percent = round(100 * n / sum(n), 1))
```

a) What number and percentage of participants smoke? Write it as n / N (%).

Your answer

b) What percentage have tertiary education? State your denominator and why.

Your answer

Exercise 1.5. Centre and spread

Calculate the mean, SD, median and IQR of `sbp_0` and of `crp` (remember `na.rm = TRUE`).

a) Fill in: SBP mean ____ (SD ____), median ____ (IQR ____ to ____). CRP mean ____ (SD ____), median ____ (IQR ____ to ____).

Your answer

b) Which summary would you report for each variable, and why?

Your answer

Exercise 1.6. Plots

Make a histogram of `bmi` and a boxplot of `crp` by `smoker`. Try adding `scale_y_log10()` to the boxplot.

```
ggplot(hearts, aes(x = bmi)) + geom_histogram(binwidth = 1)
ggplot(hearts, aes(x = smoker, y = crp)) + geom_boxplot() + scale_y_log10()
```

Describe the shape of the BMI distribution, and what the boxplot suggests about CRP in smokers and non-smokers. Why is the log scale helpful here?

Your answer

Part C1. Compare groups (script 03_compare_groups.R, 35 minutes)

For each comparison, first write the question, the outcome type, the groups, and whether the data are independent or paired.

Exercise 1.7. Two independent groups

Does mean BMI differ between women and men? Adapt the t-test code from the script.

```
t.test(bmi ~ sex, data = hearts)
```

a) Report the mean BMI in each group, the difference and its 95% CI.

Your answer

b) Interpret the result in one or two sentences. Does a p-value above 0.05 show that men and women have the same mean BMI?

Your answer

Exercise 1.8. Paired data

Did systolic blood pressure change between baseline and 12 months in the Standard advice group?

```
std <- hearts |> filter(programme == "Standard advice", !is.na(sbp_12))
t.test(std$sbp_12, std$sbp_0, paired = TRUE)
```

a) Report the mean change and 95% CI. How many people does it use?

Your answer

b) Compare with the Lifestyle programme group in the script (-6.0 mmHg). Why can we not judge the programme from the within-group change alone? What comparison does?

Your answer

Exercise 1.9. More than two groups

Does mean BMI differ between the four districts? Summarise by district, then use Welch's ANOVA and Holm-adjusted pairwise comparisons.

```
oneway.test(bmi ~ district, data = hearts)
pairwise.t.test(hearts$bmi, hearts$district, pool.sd = FALSE, p.adjust.method = "holm")
```

a) What does the overall test tell you? What does it not tell you?

Your answer

b) Which districts appear to differ? Why adjust the pairwise p-values?

Your answer

Part C2. Categorical associations (script 04_categorical.R, 20 minutes)

Exercise 1.10. A 2 x 2 table

Is high blood pressure at baseline (`high_bp_0`) associated with sex?

```
tab_sex <- table(hearts$sex, hearts$high_bp_0)
addmargins(tab_sex)
prop.table(tab_sex, margin = 1)
chisq.test(tab_sex)
```

a) Write the proportion with high BP in each sex as n / N (%).

Your answer

b) Calculate the prevalence ratio and the odds ratio (men compared with women). Why are they different? Which word should you use: risk or prevalence? Why?

Your answer

c) Interpret the chi-square test.

Your answer

Exercise 1.11. A larger table

Does the prevalence of high BP differ between districts? Make a 4 x 2 table, row proportions and a chi-square test.

Which districts have the highest and lowest prevalence? What does the chi-square test tell you, and what does it not tell you?

Your answer

Day 1 extensions (optional)

Extension E1.1. Kruskal-Wallis

Compare `activity_min` across education levels with medians (IQR) and `kruskal.test()`. What does the test compare? Is it a test of medians?

Extension E1.2. An ordinal outcome

Compare `satisfaction` between women and men. Show the percentage at each level, then use `wilcox.test()`. Why is a mean less suitable here?

Extension E1.3. Student or Welch?

Compare `summary(aov(sbp_0 ~ district, data = hearts))` with `oneway.test(sbp_0 ~ district, data = hearts)`. When would the two disagree?

Extension E1.4. McNemar's test

In the Standard advice group, cross-tabulate `high_bp_0` and `high_bp_12` and run `mcnemar.test()`. Which cells matter? Compare with the Lifestyle group in the script.

Extension E1.5. Confounding by age

Create `age_group` (30-49, 50-74) with `cut(age, c(29, 49, 74))`. Estimate the smoking-high BP OR in each age group and the Mantel-Haenszel OR (`mantelhaen.test()`). Compare with the crude OR of 3.39. What do you conclude?

Extension E1.6. Small counts

In the script, look at adverse events among participants aged 65 and over. Why is Fisher's exact test preferred here, and how would you describe the result?

Day 2 practical: regression and an introduction to data science

Part A. Correlation and linear regression (script 05_correlation_regression.R, 50 minutes)

Exercise 2.1. Scatterplot and correlation

Plot `sbp_0` against `bmi` and calculate Pearson's and Spearman's correlations.

```
ggplot(hearts, aes(x = bmi, y = sbp_0)) + geom_point(alpha = 0.4)
cor.test(~ sbp_0 + bmi, data = hearts)
cor.test(~ sbp_0 + bmi, data = hearts, method = "spearman", exact = FALSE)
```

Describe the relationship (direction, form, strength). Report r with its 95% CI. Does this show that BMI raises blood pressure?

Your answer

Exercise 2.2. Simple linear regression

Fit `lm(sbp_0 ~ bmi, data = hearts)`. Then create `bmi5 = bmi / 5` and refit.

```
hearts <- hearts |> mutate(bmi5 = bmi / 5)
fit_bmi <- lm(sbp_0 ~ bmi, data = hearts)
tidy(fit_bmi, conf.int = TRUE); glance(fit_bmi)
par(mfrow = c(2, 2)); plot(fit_bmi); par(mfrow = c(1, 1))
```

a) Interpret the slope per 1 kg/m2 and per 5 kg/m2, with 95% CIs.

Your answer

b) What is R-squared and what does it mean here?

Your answer

c) What do the residual plots show? Are the assumptions reasonable?

Your answer

Exercise 2.3. Confounding

Is smoking associated with SBP? Fit a crude model and one adjusted for age and sex.

```
fit_s0 <- lm(sbp_0 ~ smoker, data = hearts)
fit_s1 <- lm(sbp_0 ~ smoker + age + sex, data = hearts)
tidy(fit_s0, conf.int = TRUE); tidy(fit_s1, conf.int = TRUE)
```

a) Report the crude and adjusted differences in SBP for smokers. Why did the estimate change?

Your answer

b) Interpret the adjusted coefficient in one sentence starting "Among people of the same ...".

Your answer

c) A colleague suggests also adjusting for CRP, which smoking raises. Is CRP a confounder or more like a mediator here? What would adjusting for it do? Draw a small diagram.

Your answer

Exercise 2.4. Plot a fitted relationship

Fit `lm(sbp_0 ~ bmi + age + sex, data = hearts)` and plot predicted SBP against BMI for women and men aged 50, following section E of the script.

What does the plot show? What does the model assume about the BMI slope in women and men?

Your answer

Part B. Logistic and Poisson regression (script 06_logistic_poisson.R, 20 minutes, guided)

Exercise 2.5. Logistic regression

Model high BP at baseline using BMI (per 5 kg/m2), age (per 10 years) and sex.

```
hearts <- hearts |> mutate(age10 = age / 10, bmi5 = bmi / 5)
fit_lb <- glm(high_bp_0 ~ bmi5 + age10 + sex, family = binomial, data = hearts)
tidy(fit_lb, conf.int = TRUE, exponentiate = TRUE)
```

a) Interpret the aOR for `bmi5` with its 95% CI.

Your answer

b) Why should this aOR not be described as "times the risk"?

Your answer

Exercise 2.6. Poisson regression

Compare the rate of unplanned clinic visits between smokers and non-smokers, adjusting for age, using an offset for follow-up time.

```
fit_ps <- glm(clinic_visits ~ smoker + age10 + offset(log(followup_months)),
              family = poisson, data = hearts)
tidy(fit_ps, conf.int = TRUE, exponentiate = TRUE)
sum(residuals(fit_ps, type = "pearson")^2) / df.residual(fit_ps)
tidy(update(fit_ps, family = quasipoisson), conf.int = TRUE, exponentiate = TRUE)
```

a) Interpret the IRR for smoking. Why is the offset needed?

Your answer

b) What does the dispersion ratio show, and how does the quasi-Poisson CI differ?

Your answer

Part C. Clustering (script 07_clustering.R, 20 minutes, guided)

Exercise 2.7. k-means: two or three clusters?

Run the script, then fit k-means with k = 2 on the same scaled variables and describe both solutions.

```
set.seed(2024)
km2 <- kmeans(life_scaled, centers = 2, nstart = 25)
life$k2 <- factor(km2$cluster)
life |> group_by(k2) |> summarise(n = n(), act = median(activity_min),
                                  diet = mean(diet_score), bmi = mean(bmi))
table(life$k2, life$cluster)
```

a) Give each cluster a short descriptive name (k = 2 and k = 3).

Your answer

b) Why were the variables scaled? What happened without scaling?

Your answer

c) Give two reasons to be cautious before reporting these clusters as real groups.

Your answer

Day 2 extensions (optional)

Extension E2.1. Analysing a randomised trial

Compare the programme effect estimated from change scores (Welch t-test on `sbp_12 - sbp_0`) with the baseline-adjusted estimate `lm(sbp_12 ~ programme + sbp_0)`. Which CI is narrower?

Extension E2.2. Train and test

Using the split in section G of script 05, fit `lm(sbp_0 ~ age + sex + bmi)` on the training data and compare training and test RMSE. Why is test error the honest measure? Give one example of data leakage.

Extension E2.3. Negative binomial

Fit the clinic visits model with `MASS::glm.nb()` (script 06). Compare the IRR and CI with Poisson and quasi-Poisson.

Extension E2.4. Linkage methods

Repeat the hierarchical clustering with `method = "average"` and cut at 3 groups. Compare the group sizes with Ward's method. What does this tell you?

Final exercise: choose, plot, conclude (25 minutes)

Work alone or in pairs. Pick ONE question. Use the cheat sheet and decision diagrams, and the template script `R/08_final_exercise.R`. There is more than one defensible analysis for each question; what matters is that you can justify yours.

	Question
F1	Were participants in the lifestyle programme more satisfied with their care at 12 months than those given standard advice?
F2	Do men and women differ in C-reactive protein (CRP) at baseline?
F3	Is a healthier diet score associated with lower baseline systolic blood pressure, after allowing for age and sex?
F4	Was high blood pressure at 12 months less common in the lifestyle programme group than in the standard advice group?

Step 1. Write your question and the outcome, its type, the groups or exposure, and the design (independent or paired; any confounders).

Your answer

Step 2. Which analysis did you choose, and why? Name one alternative and say why you did not choose it.

Your answer

Step 3. Describe the data: give the numbers you will report (with denominators or n).

Your answer

Step 4. Produce one plot that shows the answer. Paste it here or sketch it.

Plot

Step 5. Results: the effect estimate with its 95% CI (and p-value if you wish). What did you check?

Your answer

Step 6. Conclusion in plain language (3 to 5 sentences) for a reader with no statistical training. Include the size of the effect, the uncertainty, whether it matters in practice, and one limitation.

Conclusion

Reference section

The pages below are the same as the stand-alone reference pack in the `reference` folder (cheat sheet, decision diagrams and method-selection guide). Keep them for use after the course.

Statistics and R cheat sheet

Statistics and R cheat sheet

Concepts of Biostatistics and Data Science: a two-day crash course with practical analysis in R

All examples use the course dataset `hearts` (the Healthy Hearts District Study), loaded with `hearts <- readRDS("data/hearts.rds")`. The data are **simulated for teaching**: the results illustrate methods and say nothing about real people. Packages: `readr`, `dplyr`, `ggplot2`, `broom`; `MASS` and `cluster` come with R. For choosing a method, use the decision diagrams D0 to D6.

Before any analysis: five questions

1. **Question.** Describe, compare groups, assess an association, predict, or discover groups?
2. **Outcome type.** Continuous, binary, nominal, ordinal, count or time-to-event?
3. **Design.** Independent, paired, repeated or clustered observations?
4. **Groups or predictors.** Two groups, more than two, one predictor, several?
5. **Effect to report.** Difference, ratio or slope, with its 95% CI. A p-value alone is not an answer.

Reading results honestly

- **Statistical significance is not practical importance.** A tiny difference can be "significant" in a large study. Judge size from the estimate and CI.
- **Significance is not causation.** In observational data, confounding, bias and chance can all produce associations.
- **"Not significant" is not "no effect".** Look at the CI: it may include both no effect and important effects.
- A p-value is not the probability that the null hypothesis is true.

Glossary

Observation one unit measured (here, one participant; one row).

Variable a characteristic recorded for each observation (one column).

Population, sample the people we want to learn about; the subset we measured.

Estimate a number computed from the sample to describe the population (mean difference, OR, slope).

Standard error (SE) how much an estimate would vary from sample to sample.

95% confidence interval (CI) a range of population values compatible with the data. Across many studies, 95% of such intervals would contain the true value.

Null hypothesis a reference claim, usually "no difference" or "no association".

p-value how surprising data at least this extreme would be if the null hypothesis were true.

Effect size the size of a difference or association, in units people understand.

Independent observations no observation tells you anything about another (no pairing, repeats or clusters).

Paired two measurements linked to the same person or a matched pair.

Residual observed value minus the value fitted by a model.

Rank position of a value when all values are sorted (1 = smallest).

Skewed a distribution with a long tail on one side (CRP has a long right tail).

Confounder a common cause of the exposure and the outcome. Adjust for it.

Mediator a variable on the path from exposure to outcome. Do not adjust for it if you want the total effect.

Effect modification the association differs between subgroups (e.g. women and men).

Odds number with the outcome divided by number without it.

Offset a term that tells a count model how long each person was observed.

Overdispersion counts vary more than a Poisson model assumes.

Censoring the event has not (yet) been observed when follow-up ends.

Scaling (standardising) subtract the mean and divide by the SD, so variables share a scale.

Describing data

Variable	Summary	Plot
Binary, nominal	n and % (state the denominator)	bar chart
Ordinal	n and % in order; median (IQR)	ordered bars
Continuous, symmetric	mean (SD)	histogram, boxplot
Continuous, skewed	median (IQR)	histogram (log scale)
Count	mean, variance; rate per person-time	bar chart
Two categorical	table with row %	bars of proportions
Continuous by group	summary per group	boxplot with points
Two continuous	correlation	scatterplot

```
count(hearts, district) |> mutate(pct = 100 * n / sum(n))
hearts |> group_by(sex) |> summarise(n = n(),
  mean = mean(sbp_0), sd = sd(sbp_0))
median(hearts$crp, na.rm = TRUE)
ggplot(hearts, aes(x = district, y = sbp_0)) +
  geom_boxplot() + geom_jitter(width = 0.15)
```

Course data: SBP mean 129.3 (SD 15.6) mmHg, n = 480. CRP is right-skewed: median 1.9 (IQR 1.2 to 3.7) mg/L, n = 466 (14 missing); its mean (3.0) is pulled up by a few high values.

Comparing a continuous or ordinal outcome

Welch t-test: two independent groups

Question Do two groups differ in their mean outcome?

Data Continuous outcome; two independent groups.

Assumptions Independent observations; group means approximately normal (usually fine in moderate or large groups unless there is strong skew or outliers). Welch does not assume equal variances, so it is a sensible default. Check: boxplots or histograms by group.

R `t.test(sbp_0 ~ sex, data = hearts)`

Report Difference in means with 95% CI.

Course Women 126.4, men 132.5 mmHg. Difference (women minus men) −6.1 mmHg (95% CI −8.9 to −3.4).

In words On average women's SBP was about 6 mmHg lower; the data are compatible with a difference of about 3 to 9 mmHg.

Mistake Ignoring which group is subtracted (R uses the factor level order), or reporting only the p-value.

Paired t-test: the same people measured twice

Question Did the mean change within people (or within matched pairs)?

Data Continuous outcome measured twice on the same unit.

Assumptions Pairs independent of each other; differences approximately normal or enough pairs. Check: histogram of the differences.

R `t.test(prog$sbp_12, prog$sbp_0, paired = TRUE)`

Report Mean difference with 95% CI.

Course Lifestyle group (n = 224): mean change −6.0 mmHg (95% CI −7.2 to −4.8).

Mistake Analysing paired data as independent groups (CI widens to −9.1 to −2.9). And a within-group change is not the programme effect: compare the change between the randomised groups (4.0 mmHg greater fall, 95% CI 2.3 to 5.6).

Wilcoxon rank-sum (Mann-Whitney) test

Question Do values tend to be higher in one group than in the other?

Data Ordinal or continuous outcome; two independent groups.

Assumptions Independent observations. Reading it as a shift in location needs distributions of similar shape. It is **not assumption-free** and **not in general a test of medians**.

R
```
wilcox.test(crp ~ smoker, data = hearts,
            conf.int = TRUE)
```

Report Medians (IQR) per group, and the Hodges-Lehmann shift with CI (the median of all smoker versus non-smoker differences).

Course CRP median 1.8 (IQR 1.1 to 3.3) in non-smokers, 2.8 (1.7 to 5.3) in smokers; shift 0.9 mg/L (95% CI 0.5 to 1.4), $p < 0.001$. Alternative for a positive skewed outcome: t-test on `log(crp)` gives a ratio of geometric means of 1.63 (1.31 to 2.03).

Mistake Choosing it because "the sample is small" or because a normality test gave $p < 0.05$. Choose by the question and the data.

Wilcoxon signed-rank test: paired differences

Question Do the within-pair differences tend to be above or below zero?

Assumptions Independent pairs; for the pseudomedian estimate, differences roughly symmetric.

R
```
wilcox.test(prog$sbp_12, prog$sbp_0,
            paired = TRUE, conf.int = TRUE)
```

Course Pseudomedian change −6.0 mmHg (95% CI −7.0 to −5.0).

Mistake Using the rank-sum test (independent groups) on paired data.

One-way ANOVA and Welch's ANOVA: three or more groups

Question Do mean outcomes differ across three or more independent groups?

Assumptions Independent observations; approximately normal residuals; classical ANOVA also assumes equal variances, Welch's does not. Check: `plot(fit, which = 1:2)`.

R
```
oneway.test(sbp_0 ~ district, data = hearts)
fit <- aov(sbp_0 ~ district, data = hearts)
TukeyHSD(fit)
```

Report Group means with CIs; pairwise differences with adjusted CIs if needed.

Course Means: North 126, South 134, East 131, West 126 mmHg. Welch $F = 6.83$, $p < 0.001$. Tukey: South minus North 7.6 mmHg (2.5 to 12.8).

Mistake Concluding from the overall p-value **which** groups differ, or running many unadjusted t-tests.

Kruskal-Wallis test

Question Do values tend to differ across three or more independent groups (rank-based)?

Assumptions Independent observations. Not a test of medians in general.

R `kruskal.test(crp ~ district, data = hearts)`

Course Median CRP 1.8 to 2.5 mg/L across districts; p = 0.067. Weak evidence of a difference, which is not evidence of no difference.

Mistake Following up with unadjusted pairwise tests.

Categorical outcomes

Chi-square test of association

Question Is a categorical outcome associated with a categorical exposure (do the proportions differ)?

Data Two categorical variables; independent observations.

Assumptions Expected counts about 5 or more in each cell. Check: `chisq.test(tab)$expected`.

R
```
tab <- table(hearts$smoker, hearts$high_bp_0)
prop.table(tab, margin = 1)
chisq.test(tab)
```

Report Proportions in each group, and a difference or ratio with 95% CI (see the measures section).

Course High BP in 48% of smokers and 21% of non-smokers; χ^2 = 23.1, 1 df, p < 0.001.

Mistake Reporting only the p-value; using column percentages when the question needs row percentages; using it on paired data.

Fisher's exact test

Question As for chi-square, when expected counts are small.

R `fisher.test(tab_ae)`

Course Adverse events 14/240 with the programme vs 3/240 with standard advice; p = 0.011; RR 4.67 (95% CI 1.36 to 16.03). The wide CI shows how uncertain the size of the increase is.

Mistake Thinking a small total sample decides it. The expected counts matter.

McNemar's test: paired binary data

Question Did the proportion with an outcome change within the same people?

R
```
tab_paired <- table(prog$high_bp_0,
prog$high_bp_12)
mcnemar.test(tab_paired)
```

Course Lifestyle group: 28 people moved from high BP to not high, 6 the other way; p < 0.001. Only these discordant pairs carry information about change.

Mistake A chi-square test on before and after as if they were different people.

Proportions, risks, odds and their ratios

Proportion = number with the characteristic / number in the group. Always say what the denominator is.

Risk = number who *develop* the outcome during a period / number at risk at the start. Needs follow-up (cohort study or trial).

Prevalence = number who *have* the outcome at one time / number observed then (cross-sectional study).

Odds = number with the outcome / number without it. A risk of 0.20 is odds of 0.20/0.80 = 0.25.

Risk difference (RD) = risk in exposed − risk in unexposed. Absolute; report in percentage points.

Risk ratio (RR) = risk in exposed / risk in unexposed. Relative. The same for prevalences gives a **prevalence ratio (PR)**.

Odds ratio (OR) = odds in exposed / odds in unexposed. From logistic regression with several predictors it is an **adjusted OR (aOR)**.

A value of 1 for a ratio, or 0 for a difference, means no association.

Cross-sectional example (baseline): smoking and high BP

	High BP	No high BP	Total
Smokers	38	41	79
Non-smokers	86	315	401

Prevalence: 38/79 = 0.481 vs 86/401 = 0.214

Prevalence difference: 0.267 (26.7 percentage points)

PR = 0.481 / 0.214 = **2.24** (95% CI 1.67 to 3.02)

Odds: 38/41 = 0.927 vs 86/315 = 0.273

OR = 0.927 / 0.273 = **3.39** (95% CI 2.06 to 5.61)

Adjusted for age and sex (logistic regression): aOR **2.04** (1.17 to 3.54)

Randomised trial example: adverse events

	Event	No event	Total
Lifestyle programme	14	226	240
Standard advice	3	237	240

Risk: 14/240 = 5.8% vs 3/240 = 1.25%; RD 4.6 percentage points

RR = **4.67** (95% CI 1.36 to 16.03); OR = 4.89 (1.39 to 17.26)

The outcome is rare, so OR and RR are close. For high BP (21% to 48%) the OR (3.39) is much further from 1 than the PR (2.24).

Which measures the design allows

Cohort, trial	Risks, RD, RR (and OR)
Cross-sectional	Prevalences, prevalence difference, PR (and OR)
Case-control	**Only the OR.** The proportion of cases is fixed by the design, so risks cannot be estimated.

Common mistakes

- Calling an OR of 3.4 "3.4 times the risk". That is only roughly true for rare outcomes.
- Reporting a relative measure without the absolute risks behind it.
- Mixing up percent and percentage points.

Confounding, mediation and effect modification

Confounder: a common cause of exposure and outcome. Older people in the course data are less active and have higher SBP, so age confounds the activity and SBP association. *Adjust* for confounders.

Mediator: lies on the path from exposure to outcome (activity lowers BMI, BMI raises SBP). Adjusting for it removes part of the effect you want. Do not adjust for mediators when estimating a total effect.

Effect modification: the association differs between subgroups. Report the subgroup estimates. In the course data the smoking OR was 2.75 in women and 3.28 in men: similar, so a single adjusted estimate is reasonable.

Choose adjustment variables from subject knowledge (draw a diagram of the causes), not from p-values or stepwise selection.

SBP per extra hour/week of activity (mmHg)	
Crude	−2.08 (−2.62 to −1.53)
Adjusted for age, sex (confounders)	−1.60 (−2.08 to −1.12)
Also adjusted for BMI (mediator)	−0.86 (−1.36 to −0.37)
High BP, smokers vs non-smokers	
Crude OR	3.39 (2.06 to 5.61)
OR adjusted for sex (Mantel-Haenszel)	3.09 (1.85 to 5.15)
aOR adjusted for age and sex	2.04 (1.17 to 3.54)

The change from crude to adjusted shows the confounding by age and sex. Adjustment can only remove confounding by variables that were measured, and measured well.

Correlation and regression

Pearson and Spearman correlation

Question How strongly do two variables move together?

Data Two continuous (Spearman: also ordinal) variables; independent observations.

Assumptions Pearson: a linear pattern; sensitive to outliers. Spearman uses ranks and measures a monotonic pattern. Always look at the scatterplot.

R
```
cor.test(~ sbp_0 + age, data = hearts)
cor.test(~ crp + activity_hr, data = hearts,
         method = "spearman", exact = FALSE)
```

Course Age and SBP: r = 0.50 (95% CI 0.43 to 0.57). CRP and activity: Spearman rho = −0.15.

Mistake Reading r as the size of the change (use regression), or r near 0 as "no relationship" (it could be curved).

Simple linear regression

Question How much does the mean outcome change per unit of x?

Assumptions Independence (from the design); linearity; constant spread of residuals; approximately normal residuals (matters least in large samples). The outcome itself need not be normal. Check: `plot(fit)`.

R
```
fit <- lm(sbp_0 ~ age10, data = hearts)
tidy(fit, conf.int = TRUE)
```

Report Slope with 95% CI; R^2 if useful.

Course Mean SBP 6.6 mmHg higher per 10 years of age (95% CI 5.6 to 7.6); R^2 = 0.25.

Mistake Interpreting the intercept at age 0, or predicting outside the range of the data.

Multiple linear regression

Question What is the association of x with the outcome among people with the same values of the other predictors?

R
```
fit <- lm(sbp_0 ~ activity_hr + age + sex,
          data = hearts)
```

Report Adjusted coefficients with 95% CIs, and n used.

Course −1.60 mmHg per extra hour/week (95% CI −2.08 to −1.12), adjusted for age and sex (n = 473).

In words Among people of the same age and sex, each extra hour of weekly activity was associated with SBP about 1.6 mmHg lower on average.

Mistake Adjusting for mediators; choosing variables by p-value; not noticing that missing values change n between models.

Logistic regression: binary outcome

Question How are the odds of a binary outcome related to one or more predictors?

Assumptions Independent observations; linear on the log-odds scale for continuous predictors; enough events for the number of coefficients.

R
```
fit <- glm(high_bp_0 ~ smoker + age10 + sex,
           family = binomial, data = hearts)
tidy(fit, conf.int = TRUE, exponentiate = TRUE)
```

Report (Adjusted) odds ratios with 95% CIs; predicted probabilities help explain them.

Course Smoking aOR 2.04 (1.17 to 3.54); per 10 years of age 2.29 (1.84 to 2.87); men vs women 2.06 (1.30 to 3.30).

Mistake Reading an OR as a risk ratio when the outcome is common. R models the probability of the second factor level (here "Yes").

Poisson regression: counts and rates

Question How are event rates related to predictors?

Data Count outcome; follow-up time can differ between people.

Assumptions Independent observations; variance equal to the mean. Check overdispersion: Pearson χ^2 / residual df (about 1 is fine).

R
```
fit <- glm(clinic_visits ~ programme + age10 +
  high_bp_0 + offset(log(followup_months)),
  family = poisson, data = hearts)
```

Report Incidence rate ratios (IRR) with 95% CIs.

Course Dispersion 2.03, so Poisson CIs are too narrow. Programme IRR 0.91: Poisson CI 0.82 to 1.02, quasi-Poisson CI 0.78 to 1.06.

Mistake Comparing raw counts when follow-up differs (no offset), or ignoring overdispersion. Remedies: `family = quasipoisson` or `MASS::glm.nb()`.

Clustering (unsupervised learning)

k-means clustering

Question Are there groups of people with similar values on chosen variables? There is no outcome.

Data Numeric variables, complete cases. Scale variables measured in different units.

Choices Variables; scaling; k (elbow plot, average silhouette width, interpretability); several random starts.

R
```
z <- scale(life[, c("activity_min",
      "diet_score", "bmi")])
set.seed(2024)
km <- kmeans(z, centers = 3, nstart = 25)
```

Report How variables and k were chosen; cluster sizes; profiles on the original scale.

Course Clusters of 146, 228 and 89 people (low, middle and high activity and diet scores, with BMI falling across them). Average silhouette 0.29 for k = 3 and 0.34 for k = 2: weak structure.

Mistake Not scaling (activity minutes then dominate); treating clusters as real types; testing whether clusters differ on the variables used to build them.

Hierarchical clustering

Question As for k-means, shown as a tree (dendrogram) that can be cut at any k.

R
```
hc <- hclust(dist(z), method = "ward.D2")
plot(hc); cutree(hc, k = 3)
```

Choices Distance and linkage. Ward's method gives compact groups; average linkage gave groups of 426, 30 and 7 in the course data.

Mistake Reading the dendrogram as proof that groups exist.

R essentials

`library(dplyr)` load a package (install once with `install.packages("dplyr")`)

`raw <- read_csv("data/hearts_raw.csv")` import

`glimpse(raw); summary(raw); colSums(is.na(raw))`

`mutate(activity_min = na_if(activity_min, -99))` recode a missing-value code

`mutate(district = factor(district, levels = c(...)))` first level = reference group

`filter(hearts, programme == "Lifestyle programme")`

`select(hearts, id, sbp_0); count(hearts, smoker)`

`group_by(sex) |> summarise(mean = mean(bmi, na.rm = TRUE))`

`ggplot(data, aes(x = ..., y = ...)) + geom_point()`

`ggsave("outputs/plot.png", width = 6, height = 4)`

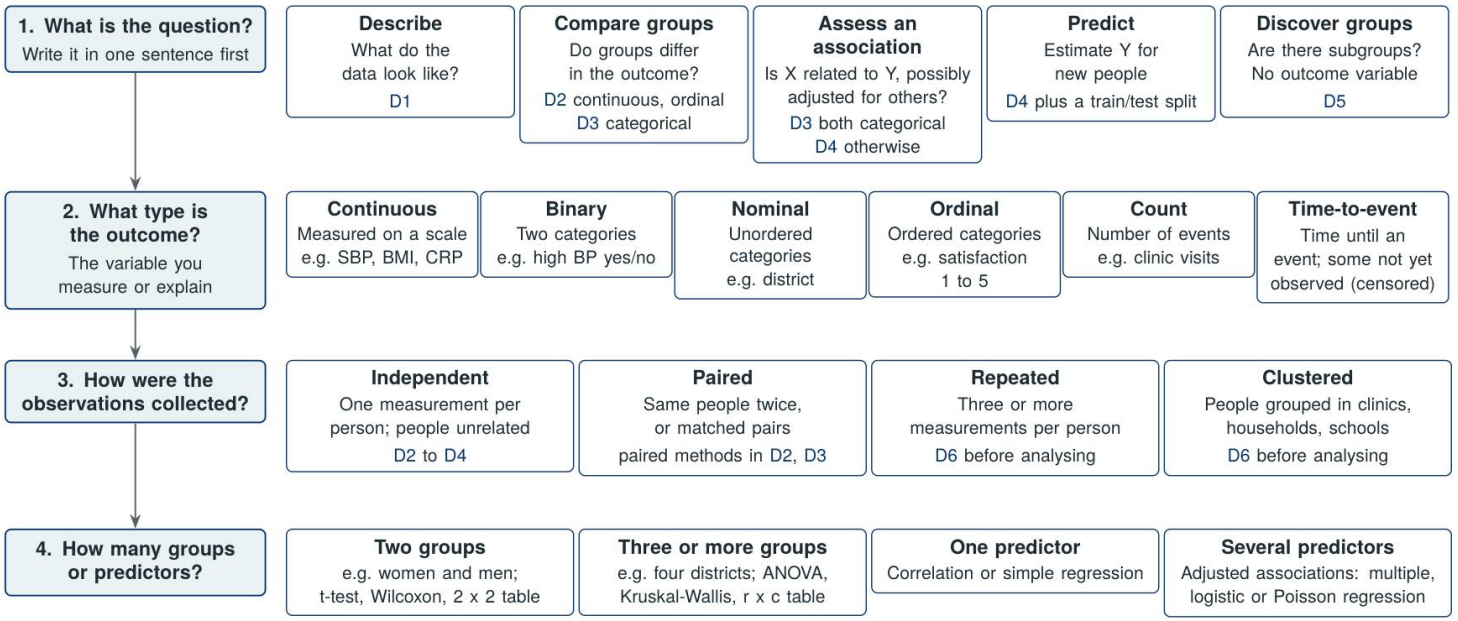
`y ~ x` a formula: outcome on the left, predictors on the right

`na.rm = TRUE` ignore missing values in a summary

`?t.test` open the help page for a function

D0. Choosing an analysis: start here

Answer four questions in order. The blue labels point to the focused diagrams D1 to D6.



Use these diagrams to structure your thinking, not as automatic rules. Look at the data with plots, think about how they were collected, and report an effect estimate (difference, ratio, slope) with its 95% confidence interval, not only a p-value. Sample size alone does not decide between a t-test and a rank-based test, and neither does the p-value of a normality test.

D1. Describing data: summaries and plots by variable type

Always report n and the number of missing values. Plot the data before choosing a summary.

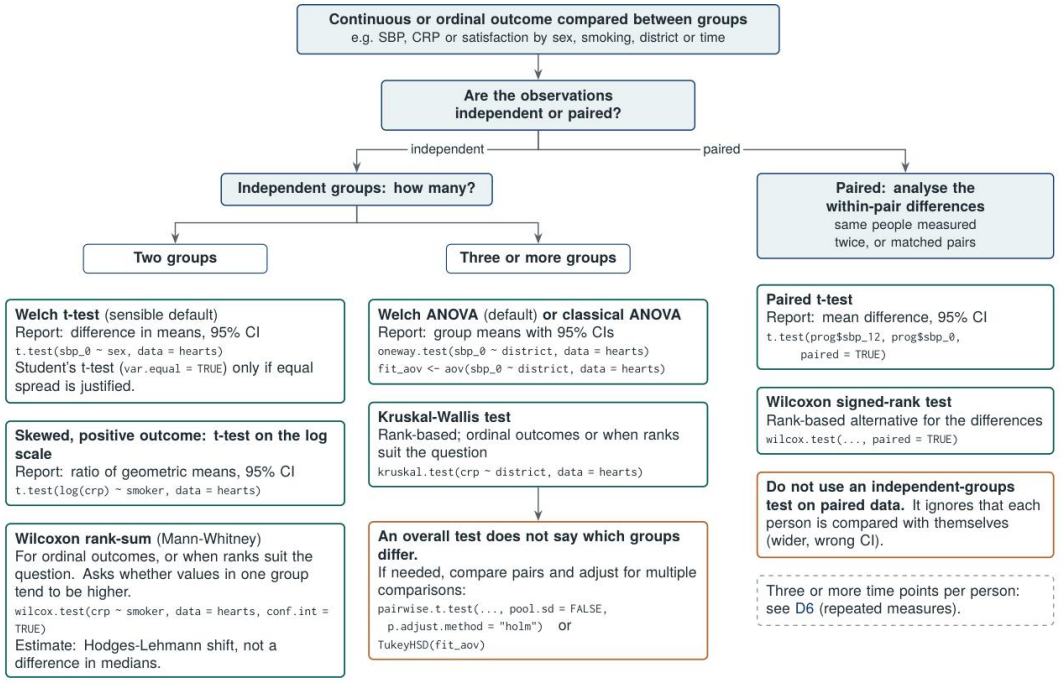
Variable type	Summarise with	Plot with	R, course data
One variable			
Binary or nominal e.g. smoker, district	Counts and percentages. State the denominator: everyone, or only those with a known value.	Bar chart of counts or percentages	`count(hearts, district)` `geom_bar()`
Ordinal e.g. satisfaction 1 to 5	Counts and percentages, kept in category order. Median and IQR can be added.	Bar chart in category order	`count(hearts, satisfaction)`
Continuous, roughly symmetric e.g. SBP	Mean and standard deviation (SD)	Histogram; boxplot	`mean(sbp_0); sd(sbp_0)` `geom_histogram(binwidth = 5)`

Continuous, skewed
e.g. CRP
	Median and interquartile range (IQR, 25th to 75th percentile). Mean and SD are pulled by extreme values.	Histogram, often on a log scale; boxplot	`median(crp, na.rm = TRUE)`
			`quantile(crp, c(0.25, 0.75), na.rm = TRUE)`
			`scale_x_log10()`

Count e.g. clinic visits	Mean and variance; a rate (events per person-time) when follow-up time differs	Bar chart of each count value	`sum(clinic_visits) / sum(followup_months)`
Two variables			
Continuous by group e.g. SBP by district	The summary above, given separately for each group, with n per group	Boxplots with the raw points	`group_by(district) |> summarise(...)` `geom_boxplot() + geom_jitter()`
Two categorical e.g. smoker by high BP	Contingency table with row percentages (denominator = each row total)	Bar chart of proportions	`tab <- table(hearts$smoker, hearts$high_bp_0)` `prop.table(tab, margin = 1)`
Two continuous e.g. age and SBP	Correlation, if the pattern is roughly linear or monotonic (see D4)	Scatterplot	`geom_point()`

D2. Comparing a continuous or ordinal outcome between groups

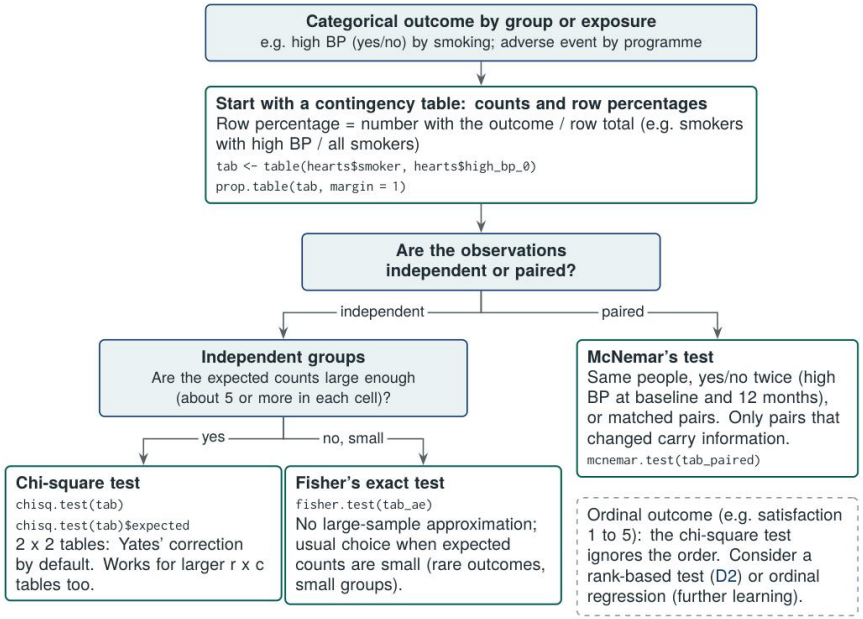
Report the size of the difference with a 95% CI. The test is only one part of the answer.



Assumptions and choices. t-tests and ANOVA assume independent observations (from the design) and approximately normal sample means. In large groups this holds even with moderate skew; with small groups, strong skew or outliers matter more. Check residual plots. Rank-based tests are not assumption-free: they also need independent observations, and they are not in general tests of medians. Do not choose a test from sample size alone or from a normality-test p-value: look at the data and decide which summary answers your question.

D3. Categorical outcomes: tables, tests and effect measures

Binary or nominal outcome compared between groups or exposures. Always state the denominator.



Which effect measure? It depends on the study design

Cohort study or randomised trial
People followed over time, so proportions are **risks**. Report risk difference (RD) and risk ratio (RR); OR also possible.
Trial example, adverse events: 14/240 (5.8%) with the programme vs 3/240 (1.2%) with standard advice. RR 4.67 (95% CI 1.36 to 16.03).

Cross-sectional study (one time point)
Proportions are **prevalences**. Report prevalence difference and prevalence ratio (PR); OR also possible.
Course example, high BP: 38/79 (48%) smokers vs 86/401 (21%) non-smokers. PR 2.24 (1.67 to 3.02); OR 3.39 (2.06 to 5.61).

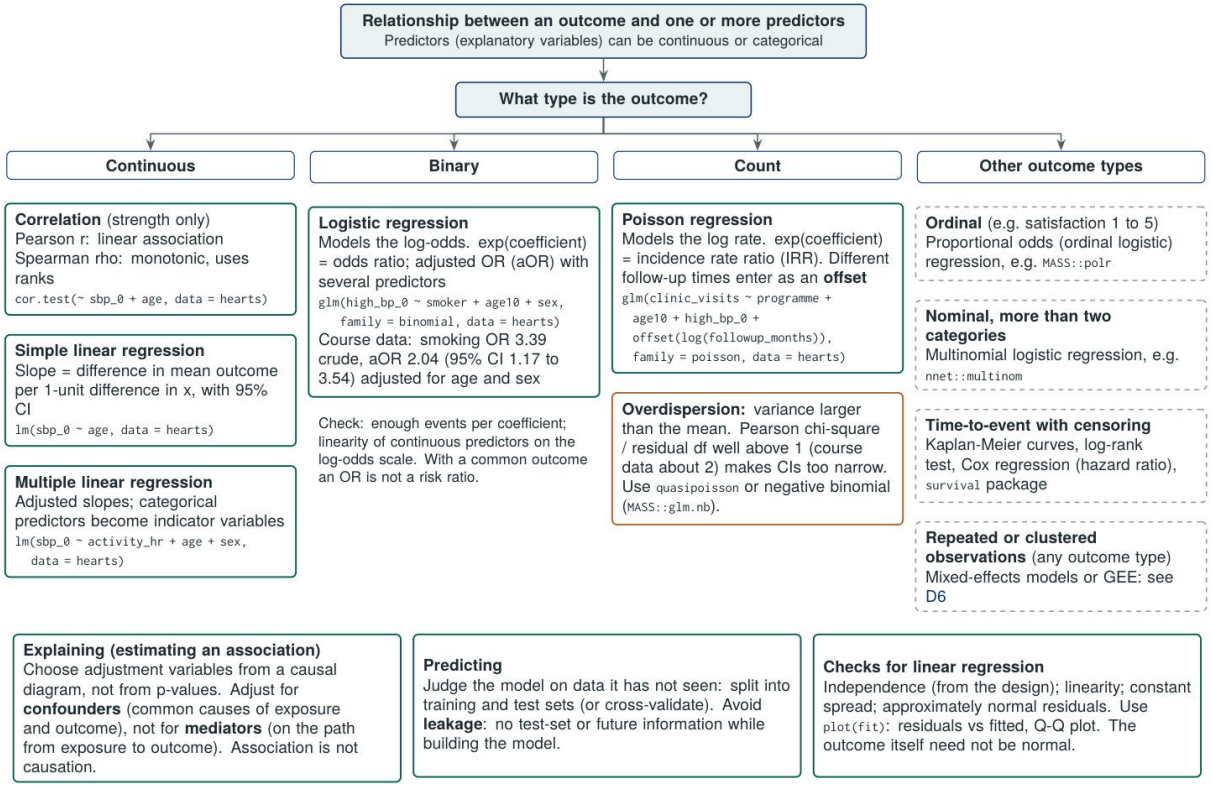
Case-control study
Cases and controls are chosen by outcome, so the proportion with the outcome is set by the design. Risks cannot be estimated: **only the OR** is interpretable.

OR is close to RR only when the outcome is rare (roughly under 10%). With a common outcome the OR is further from 1: do not read an OR of 3.4 as "3.4 times the risk".

Adjusting for a confounder
Stratify and combine: `mantelhaen.test(...)`, or logistic regression for an adjusted OR (aOR): see D4.

D4. Associations and regression models

Regression estimates how the outcome changes with each predictor, holding the other predictors constant. Dashed boxes are beyond this course: routes for further learning.



D5. Discovering groups: clustering

Unsupervised learning: there is no outcome. Clusters depend on the variables, the scaling, the method and k.

1. Purpose and variables

Write the question, e.g. "Are there distinct lifestyle profiles?"

Choose variables for substantive reasons. Every variable you add changes the distances.

Leave out outcomes you want to compare later (e.g. use activity, diet and BMI; keep SBP aside).

2. Prepare the data

Complete cases: `kmeans()` cannot handle missing values.

Very skewed variables may need a log transformation.

Variables on different scales or units?
e.g. minutes (0 to 900) and a score (0 to 10)

Yes (usual): standardise each variable to mean 0, SD 1

```
z <- scale(life[,
c("activity_min",
    "diet_score", "bmi")])
```

Without scaling, the variable with the largest numbers dominates.

3. Choose a method

k-means
Numeric variables; you set k in advance; finds compact, roughly round groups. Use several random starts.

```
set.seed(2024)
kmeans(z, centers = 3,
     nstart = 25)
```

Hierarchical clustering
Builds a tree (dendrogram) by merging the closest groups; cut it to get k groups. Choose a distance and a linkage (Ward's method is common).

```
hc <- hclust(dist(z),
     method = "ward.D2")
cutree(hc, k = 3)
```

4. Choose the number of clusters

Elbow plot: within-cluster sum of squares against k; look for a bend.

Average silhouette width: how well points sit in their own cluster (maximum 1).

Interpretability: can you describe and use the groups?

There is often no single right k. In the course data the elbow is gradual and all silhouettes are below 0.35: weak structure. Say so.

5. Check and describe

Describe clusters on the **original** scale (medians, means, n).

Check stability: other seeds, k, methods or subsets.

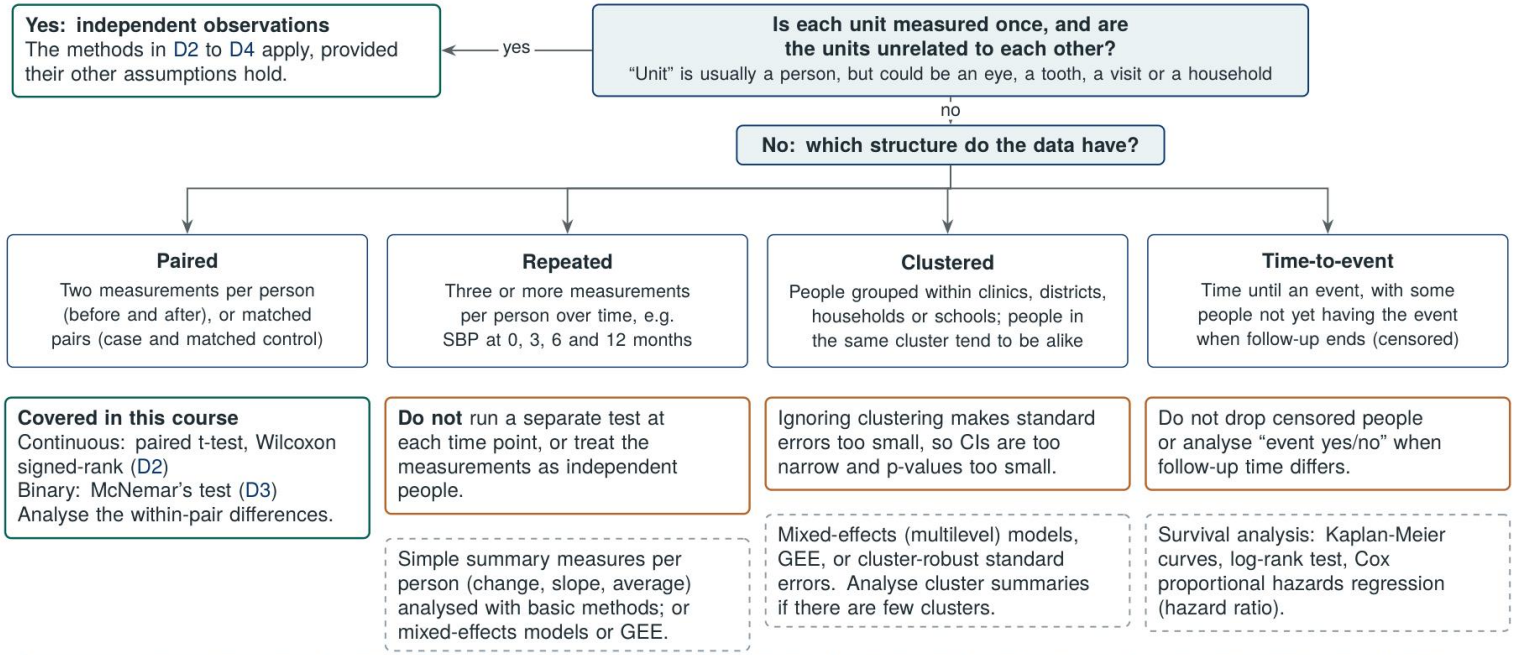
Compare clusters on variables **not** used to build them (e.g. age, SBP).

Apparent clusters are not automatically real.
k-means always returns k groups, even from data with no groups. Do not test whether clusters differ on the variables used to create them.

Further learning: mixed numeric and categorical variables (Gower distance with `cluster::daisy` and `pam`); model-based clustering (Gaussian mixture models); principal component analysis for visualising many variables.

D6. Design check: are basic independent-observation methods appropriate?

The t-tests, chi-square tests and regression models in this course assume each observation is independent of the others.



Solid teal box: methods practised in the course. Orange-brown: common mistakes. Dashed grey: further learning beyond this course. A count over different follow-up times is handled in the course with a Poisson offset (D4).

Choosing an analysis method

Concepts of Biostatistics and Data Science: a two-day crash course with practical analysis in R

Start from the **question**, then the **outcome type**, then the **design** (independent or paired), then the number of **groups or predictors**. The table suggests a usual first choice and what to report; it is a guide for thinking, not an automatic rule. The decision diagrams D0 to D6 show the same choices visually, and the cheat sheet gives assumptions, R code and common mistakes for each method. *Grey italic entries* are beyond this course and are routes for further learning.

Question	Outcome	Independent observations	Paired observations	Report	R, course example
Describe (diagram D1)					
What do the data look like?	Categorical	Counts and % (state the denominator); bar chart		n, %	`count(hearts, district)`
	Continuous	Mean (SD) if symmetric; median (IQR) if skewed; histogram, boxplot		n, missing, summary	`summarise(hearts, mean(sbp_0), sd(sbp_0))`
Compare two groups (D2, D3)					
Do two groups differ?	Continuous	Welch t-test (default); Student's t only with equal spread	Paired t-test on the differences	Mean difference, 95% CI	`t.test(sbp_0 ~ sex, data = hearts)`
	Continuous, positive, right-skewed	t-test on log scale	Paired t-test on log values	Ratio of geometric means, CI	`t.test(log(crp) ~ smoker, data = hearts)`
	Ordinal, or ranks suit the question	Wilcoxon rank-sum (Mann-Whitney)	Wilcoxon signed-rank	Medians (IQR); Hodges-Lehmann shift, CI	`wilcox.test(crp ~ smoker, data = hearts, conf.int = TRUE)`
	Binary	Chi-square; Fisher's exact when expected counts are small	McNemar's test	Proportions; RD, RR or PR, OR with CI	`chisq.test(tab)`; `fisher.test(tab_ae)`; `mcnemar.test(tab_paired)`
	Count with follow-up time	Poisson regression with offset	*conditional models*	Rate ratio (IRR), CI	`glm(clinic_visits ~ programme + offset(log(followup_months)), family = poisson, data = hearts)`
	Time-to-event	*Kaplan-Meier, log-rank test*		*Hazard ratio*	*survival package*
Compare three or more groups (D2, D3)					
Do several groups differ?	Continuous	Welch ANOVA or classical ANOVA; then adjusted pairwise comparisons if needed	*Repeated-measures models, mixed models*	Group means with CIs; adjusted pairwise differences	`oneway.test(sbp_0 ~ district, data = hearts)`; `TukeyHSD(aov(...))`
	Ordinal, or ranks	Kruskal-Wallis	*Friedman test*	Medians (IQR) per group	`kruskal.test(crp ~ district, data = hearts)`

Question	Outcome	Independent observations	Paired observations	Report	R, course example
	Categorical	Chi-square for an r x c table; Fisher when counts are small	*Cochran's Q*	Row % per group	`chisq.test(table(hearts$district, hearts$high_bp_0))`
Assess an association between two variables (D3, D4)					
Are two variables related?	Two continuous	Scatterplot; Pearson r (linear) or Spearman rho (monotonic); simple linear regression for the size of change	*Repeated or clustered: mixed models*	r with CI; slope with CI	`cor.test(~ sbp_0 + age, data = hearts)`; `lm(sbp_0 ~ age10, data = hearts)`
	Two categorical	Contingency table, chi-square or Fisher	McNemar (same binary variable twice)	RD, RR, PR or OR by design	`prop.table(tab, margin = 1)`
Estimate an association adjusted for other variables (D4)					
What is the association, holding confounders constant?	Continuous	Multiple linear regression	*Mixed models, GEE*	Adjusted slope, CI	`lm(sbp_0 ~ activity_hr + age + sex, data = hearts)`
	Binary	Logistic regression	*Conditional logistic regression (matched), GEE*	Adjusted OR (aOR), CI	`glm(high_bp_0 ~ smoker + age10 + sex, family = binomial, data = hearts)`
	Count	Poisson regression with offset; quasi-Poisson or negative binomial if overdispersed	*Mixed models, GEE*	Adjusted IRR, CI	`glm(... + offset(log(followup_months)), family = poisson, ...)`; `MASS::glm.nb()`
	Ordinal	*Proportional odds regression*		*Cumulative OR*	*MASS::polr*
	Nominal, more than 2 categories	*Multinomial logistic regression*		*Relative risk ratios*	*nnet::multinom*
	Time-to-event	*Cox proportional hazards regression*		*Hazard ratio*	*survival::coxph*
Predict an outcome for new people (D4)					
How well can we predict?	Any	The regression model for the outcome type, fitted on training data and judged on test data; no leakage		Prediction error on test data (e.g. RMSE)	`train/test split with sample()`; `predict(fit, newdata = test)`
Discover groups (D5)					
Are there subgroups?	None (no outcome)	Scale variables in different units; k-means or hierarchical clustering; check k, stability and meaning		Cluster sizes and profiles; how k was chosen	`kmeans(scale(x), centers = 3, nstart = 25)`; `hclust(dist(z), method = "ward.D2")`

When basic independent-observation methods are inappropriate (diagram D6)

Structure	Example	Problem if ignored	What to do
Paired	Same people before and after; matched case and control	Independent-groups tests give wrong (usually too wide) CIs	Analyse differences: paired t-test, Wilcoxon signed-rank, McNemar (covered in the course)
Repeated measures	SBP at 0, 3, 6 and 12 months	Separate tests at each time point inflate false positives; people counted as several independent observations	One summary per person (change, slope, mean) analysed with basic methods; *mixed-effects models or GEE*
Clustered	Patients within clinics; people within households or districts	Standard errors too small, CIs too narrow, p-values too small	*Mixed-effects (multilevel) models, GEE, cluster-robust standard errors*; with few clusters, analyse cluster summaries
Time-to-event with censoring	Time to diagnosis; some people not diagnosed by the end of follow-up	Dropping censored people or using event yes/no biases results when follow-up differs	*Kaplan-Meier, log-rank test, Cox regression*
Different follow-up times for counts	Clinic visits over 2 to 12 months	Raw counts compare unequal exposure	Poisson regression with `offset(log(followup_months))` (covered)

Principles that matter more than any flowchart

- **Sample size alone does not decide between a t-test and a rank-based test.** With small samples, look carefully at the data (strong skew, outliers, ordinal scale) and choose the summary that answers the question. Rank-based tests are not a safety net for small samples: they also have assumptions and can have little power.
- **Do not choose a test from a normality-test p-value.** In large samples such tests flag trivial departures; in small samples they miss important ones. Plots of the data and residuals are more informative. For t-tests, ANOVA and regression what matters is the distribution of the sample means or residuals, not of the raw outcome.
- **Rank-based tests are not assumption-free and not, in general, tests of medians.** Wilcoxon and Kruskal-Wallis assume independent observations and compare whole distributions through ranks.
- **A significant overall test (ANOVA, Kruskal-Wallis, r x c chi-square) does not identify which groups differ.** Use adjusted pairwise comparisons or look at the estimates.
- **Welch's versions are sensible defaults** for two-group and several-group comparisons of means, because equal variances are rarely known.
- **Independence comes from the design**, not from a test. Paired, repeated and clustered data need methods that respect their structure.
- **Report effect estimates with 95% CIs.** Statistical significance does not establish practical importance or causation, and a non-significant result does not show that there is no effect.
- **The study design determines which measures can be interpreted:** risks and RR from cohorts and trials, prevalences and PR from cross-sectional studies, only the OR from case-control studies.
- **Choose adjustment variables from subject knowledge**: adjust for confounders, not mediators.